

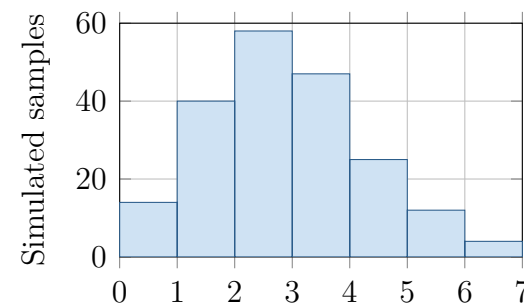
When a Normal Curve Distorts the Tail

Unit 3 · Topic 3.3 · TODAY'S PROBLEM

Suppose an app developer's model says that $p_0 = 0.20$ of all new users activate voice commands during their first day. A simple random sample of 12 new users includes 6 who activate the feature, so $\hat{p} = 0.50$. Under the developer's model, a class simulates 200 samples of 12 independent users; the histogram shows the number of activations in each sample.

Mara: "A normal model for $\hat{p}$ gives $P(\hat{p} \geq 0.50) \approx 0.005$, so that is the chance probability I would report."

Jules: "The simulated distribution is not very normal. I would estimate the chance probability directly from its tail."



Voice-command activations in one sample of 12

FIRST TAKE (5 min, on your sheet)

Which chance probability would you report, Mara's or Jules's? Cite one detail from the histogram or sample size.

- DOK 1** (a) State the most common number of activations in the simulated samples and describe the distribution as roughly symmetric, skewed left, or skewed right.
- DOK 2** (b) Use the histogram to estimate $P(\hat{p} \geq 0.50)$ under $p_0 = 0.20$. Then calculate np_0 and $n(1 - p_0)$ and assess the expected-count condition for a normal approximation.
- DOK 3** (c) Choose the better chance-probability method and use a 5% criterion to judge evidence against $p_0 = 0.20$. Cite the simulated tail and a shape or expected-count feature, compare 0.020 with 0.005, and explain the reader consequence of Mara's report.

Video + follow-along: u6_lesson1-2_live.html (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

